Exercise Sheet 7

Structural Balance

5942UE Network Science

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7.A Triadic Balance Theory

Exercise 7.A.1: Classifying Signed Triangles

Label each signed triangle as **balanced (B)** or **unbalanced (U)**:

1. $A \overset{+}{-} B, B \overset{+}{-} C, A \overset{+}{-} C$ (all positive)
2. $A \overset{+}{-} B, B \overset{+}{-} C, A \overset{-}{-} C$ (two positive, one negative)
3. $A \overset{-}{-} B, B \overset{-}{-} C, A \overset{+}{-} C$ (one positive, two negative)
4. $A \overset{-}{-} B, B \overset{-}{-} C, A \overset{-}{-} C$ (all negative)

Which pattern is most stable, and which is "socially unstable"?

Solution:

A triangle is balanced iff the product of edge signs is positive (0 or 2 negatives).

1. **(+,+,+):** Balanced. Three mutual friends — most stable.
2. **(+,+,−):** Unbalanced. Classic "two of my friends hate each other" tension.
3. **(−,−,+):** Balanced. Two allies sharing a common enemy.
4. **(−,−,−):** Unbalanced. Three mutual enemies — no coalition possible.

The (+,+,−) pattern is the most common source of social tension; (+,+,+) is the most stable.

Exercise 7.A.2: Applying the Balance Theorem

Complete signed graph on $1, 2, 3, 4, 5$. All edges positive **except** $1–3$, $1–4$, $2–3$, $2–4$, $5–3$, and $5–4$ (which are negative).

1. Check whether every triangle is balanced.
2. If balanced, find the **two-camp partition** guaranteed by the Balance Theorem.
3. Verify within-camp edges are positive and between-camp edges negative.
4. What does the partition represent socially?

Solution:

1. Examples: $1, 2, 3$: (+,−,−) → balanced. $1, 2, 4$: (+,−,−) → balanced. $1, 3, 4$: (−,−,+) → balanced. $3, 4, 5$: (+,−,−) → balanced. All triangles are balanced.

2. Partition: $X = 1, 2, 5, Y = 3, 4$.

3. Verification: All within-X edges (1–2, 1–5, 2–5) are positive; the within-Y edge 3–4 is positive; all X–Y edges are negative.

4. Interpretation: Two allied factions in mutual conflict — the textbook two-bloc geopolitical structure ([?EasleyKleinberg2010]).

7.B Weak Balance

Exercise 7.B.1: Strong vs. Weak Balance

Graph with two triangles connected by one edge:

- A, B, C all positive
- D, E, F all negative
- Connecting edge C–D is negative

1. Is A, B, C balanced under strong balance? Under weak balance?
2. Is D, E, F balanced under strong balance? Under weak balance?
3. What does the global partition look like under each?
4. In international relations, what does an all-negative triangle represent?

Solution:

1. A, B, C — all positive: balanced under both strong and weak balance.

2. D, E, F — all negative: **unbalanced** under strong balance (product is -1). Weak balance *forbids only* (+,+,−) triangles, so all-negative is **balanced** (weakly).

3. Partition:

- *Strong balance:* impossible without resolving D, E, F (one edge must flip).
- *Weak balance:* allows **more than two camps**, e.g. A, B, C, D, E, F — three mutual enemies form three separate factions.

4. Three mutually hostile countries. Strong balance theory predicts an alliance forms (resolving to two camps). Weak balance treats multi-polar rivalry as stable.

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Exercise 7.B.2: Finding Camps in a Signed Network

A signed graph: $1, 2, 3$ are mutually friends, $4, 5, 6$ are mutually friends, all cross-group edges are negative.

1. Check every triangle for balance.
2. Find the two-camp partition.
3. Verify within-group edges are positive, between-group negative.
4. Flip the cross-group edge 1–4 from negative to positive. Which triangles become unbalanced?

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Solution:

```
import networkx as nx
from itertools import combinations

G = nx.Graph()
G.add_edges_from([(1,2),(1,3),(2,3),(4,5),(4,6),(5,6)], sign=1)
G.add_edges_from([
    (1,4),(1,5),(1,6),
    (2,4),(2,5),(2,6),
    (3,4),(3,5),(3,6),
], sign=-1)

def triangle_sign(G, t):
    a, b, c = t
    return G[a][b]["sign"] * G[b][c]["sign"] * G[a][c]["sign"]

triangles = [t for t in combinations(G.nodes, 3)
             if all(G.has_edge(u, v) for u, v in combinations(t, 2))]
balanced = sum(1 for t in triangles if triangle_sign(G, t) > 0)
```

The partition $1, 2, 3$ vs. $4, 5, 6$ makes every triangle balanced. Flipping the rogue edge $1 \overset{+}{-} 4$ creates four unbalanced triangles: $1, 2, 4$, $1, 3, 4$, $1, 4, 5$, and $1, 4, 6$. Each contains exactly one negative edge. The structural tension typically resolves either by breaking the rogue tie or by realigning groups.

7.C Relaxed Balance and Applications

Exercise 7.C.1: WWI Alliance Network

Five countries during WWI:

- **Allies**: France, Britain, Russia (mutually positive)
- **Central Powers**: Germany, Austria-Hungary (mutually positive)
- All Allies–Central Powers edges are negative

1. Is this network perfectly balanced? Identify any unbalanced triangles.
2. Does it match the two-camp Balance Theorem?
3. What if a new country joins with positive ties to **both** camps?
4. Italy switched from Germany to the Allies in 1915 — what does balance theory predict?

Solution:

1. All within-camp triangles are (+,+,+). Cross-camp triangles like F, G, B are (−,−,+) — balanced. The network is perfectly balanced.

2. Yes — exactly the two-camp realisation ([?EasleyKleinberg2010]).

3. A new country with positive ties to both camps creates (+,+,−) triangles immediately — unstable. Balance theory predicts the country must eventually choose a side.

4. Italy initially had a positive tie to Germany while Allies–Germany was negative — straddling two hostile camps. The (+,+,−) tension is exactly what balance theory predicts must resolve, and the historical switch in 1915 eliminated the unbalanced triangles.

Exercise 7.C.2: Approximate Balance in Real Networks

Using the karate club graph, sign edges by faction (within = +, cross = −), then flip a small number of cross-faction edges to positive to simulate noisy or ambiguous relationships.

1. Count balanced vs. unbalanced triangles.
2. What fraction of triangles are balanced?
3. Remove edges in order of how many unbalanced triangles they participate in. After removing the top 5, how does the fraction change?
4. Plot fraction-balanced vs. edges-removed.

Solution:

```python
import networkx as nx
import matplotlib.pyplot as plt
from itertools import combinations

G = nx.karate_club_graph()
factions = nx.get_node_attributes(G, "club")
for u, v in G.edges():
    G[u][v]["sign"] = 1 if factions[u] == factions[v] else -1

# Add a small amount of sign noise: these are cross-faction ties
# that are treated as positive instead of negative.
for u, v in [(8, 30), (8, 32)]:
    G[u][v]["sign"] = 1

def triangles(G):
    return [t for t in combinations(G.nodes, 3)
            if all(G.has_edge(u, v) for u, v in combinations(t, 2))]
```

With signs derived exactly from faction labels, the karate club triangles would be perfectly balanced by construction. The deliberate sign noise creates a small number of unbalanced triangles. Removing high-impact edges — those participating in many unbalanced triangles — increases the global balance fraction, mirroring how social groups may resolve tension by severing or reclassifying conflicting ties.

Wrap-Up

- triangle sign products give a one-line balance test
- the Balance Theorem turns a local rule into a global two-camp partition
- weak balance permits more than two camps and is often more realistic
- real networks are rarely perfectly balanced — approximate balance is the typical regime